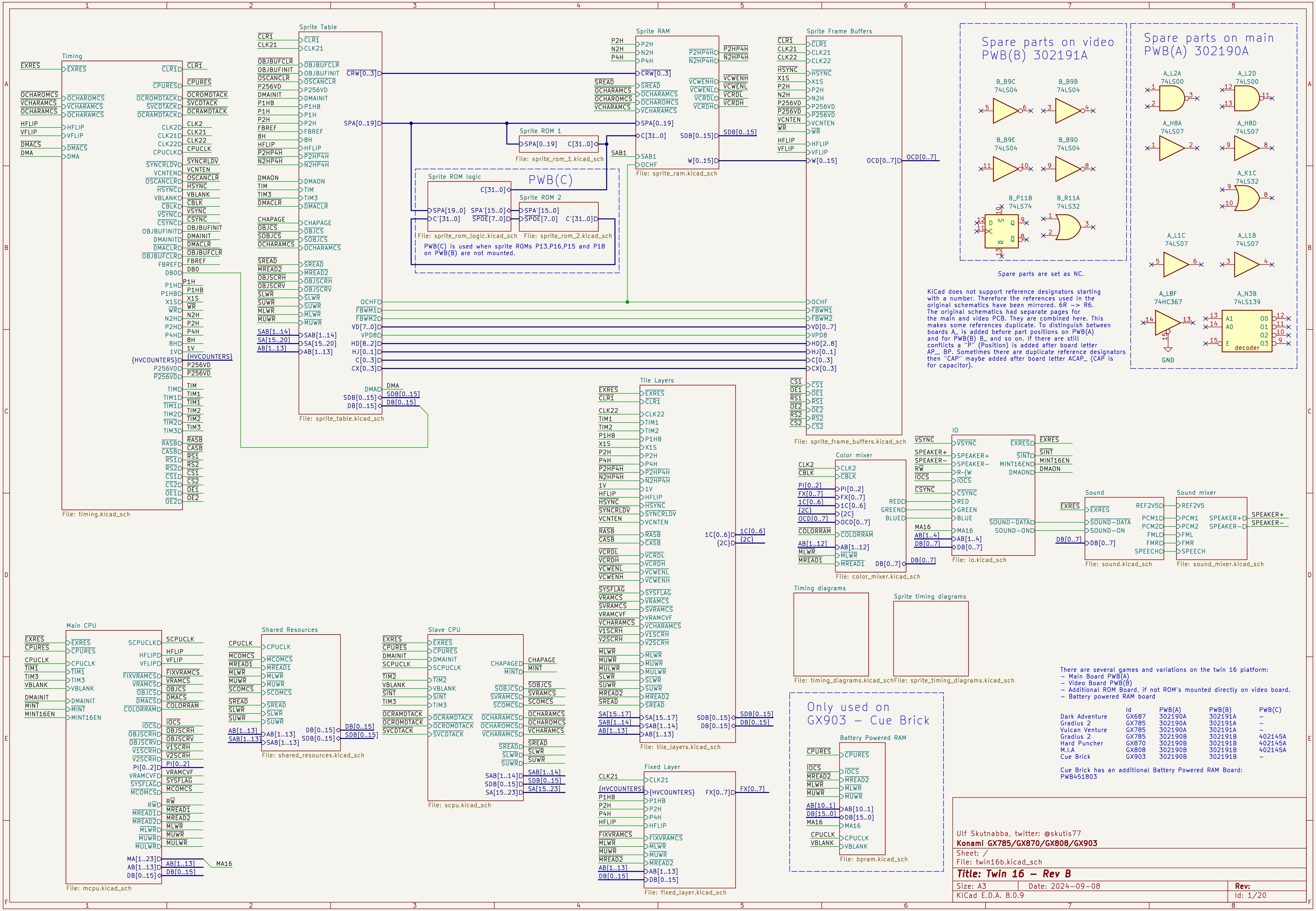
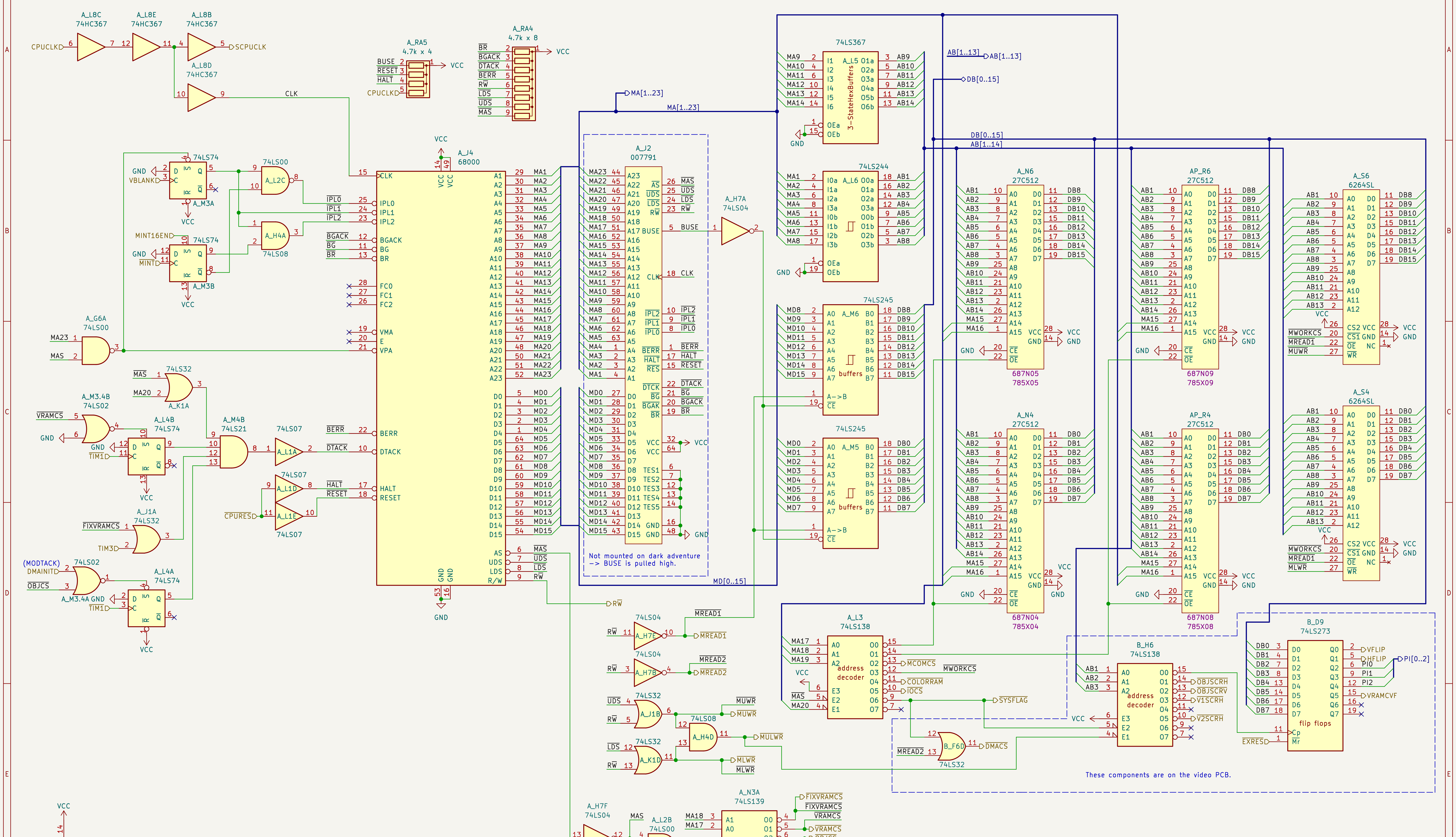


	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–



Ulf Skutnabba, twitter: @skutis77

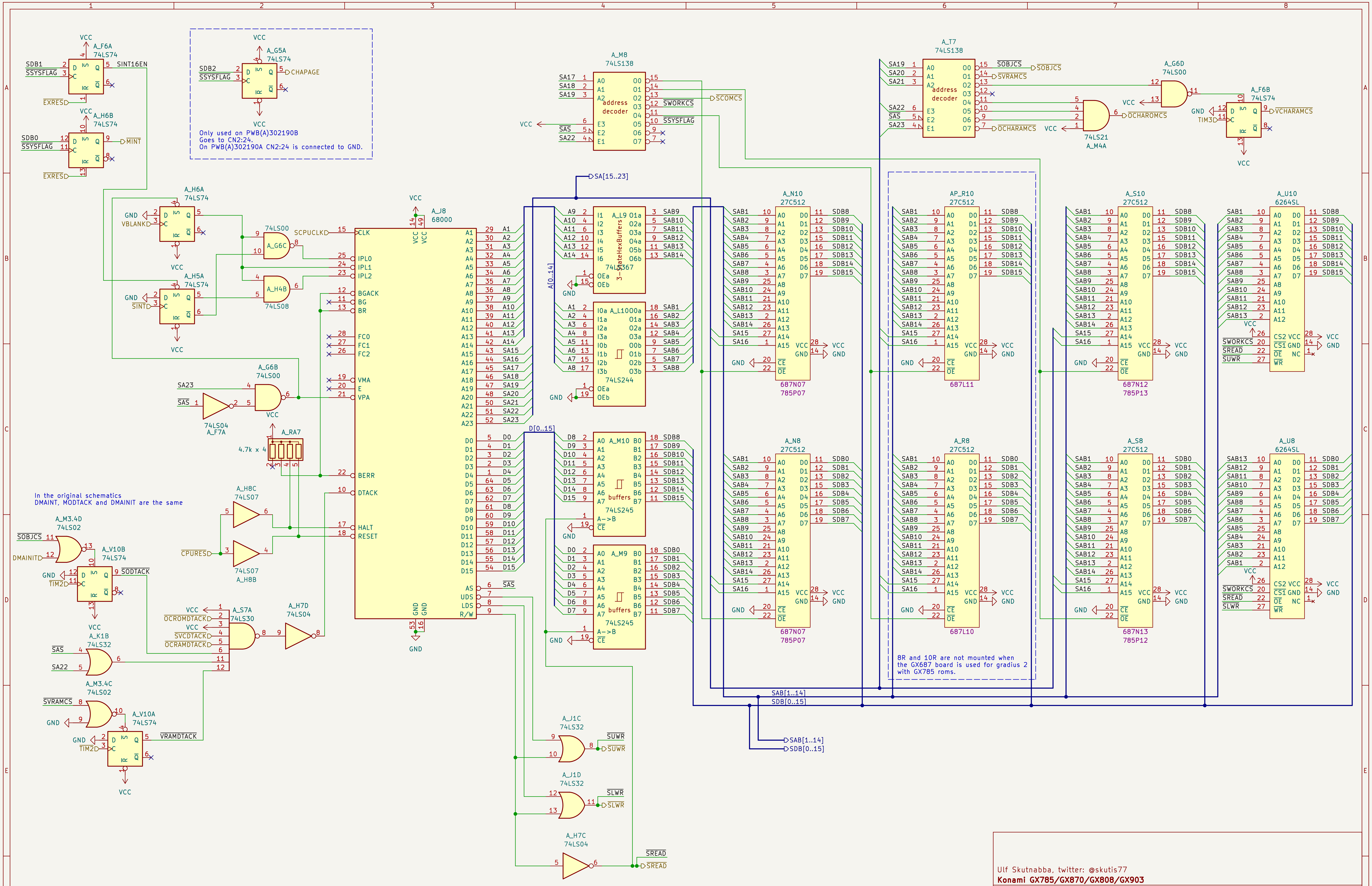
Konami GX785/GX870/GX808/GX903

Sheet: /Main CPU/

File: mcpu.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 2/20

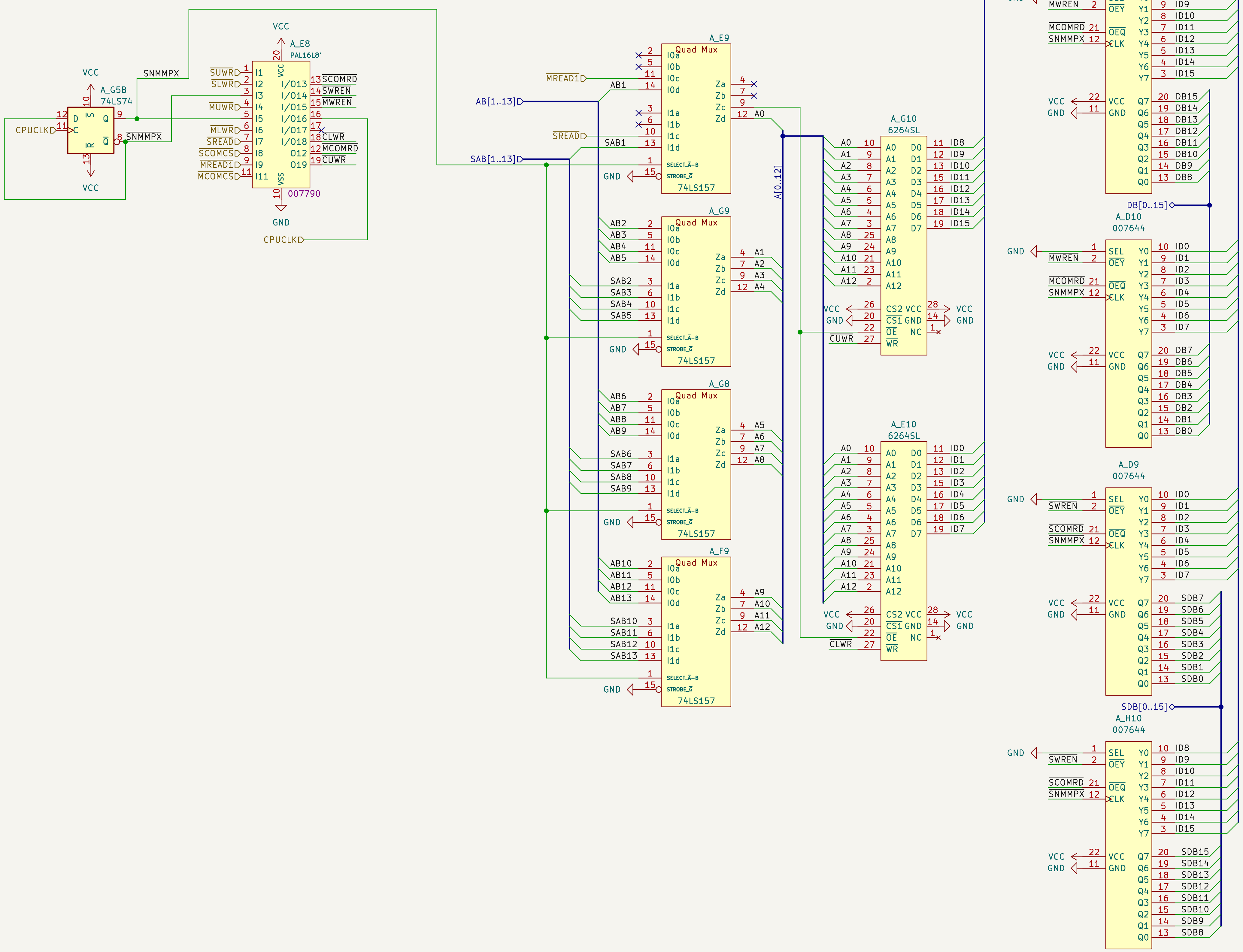


Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Slave CPU/
File: scpu.kicad_sch
Title: Twin 16 — Rev B
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 3/20

Konami 007790

```
MCOMRD = ~(~MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(~SREAD & ~SCOMCS)
SWREN = ~(~SNMMPX & SREAD & ~SCOMCS)
MWREN = ~(~SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(~SLWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(~SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)
```

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



This is from jammarcade.net

Pin 1 = SEL - changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY - Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ - Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK - Clock input

Pins 3-10 = Y outputs - normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13-20 = Q outputs - clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

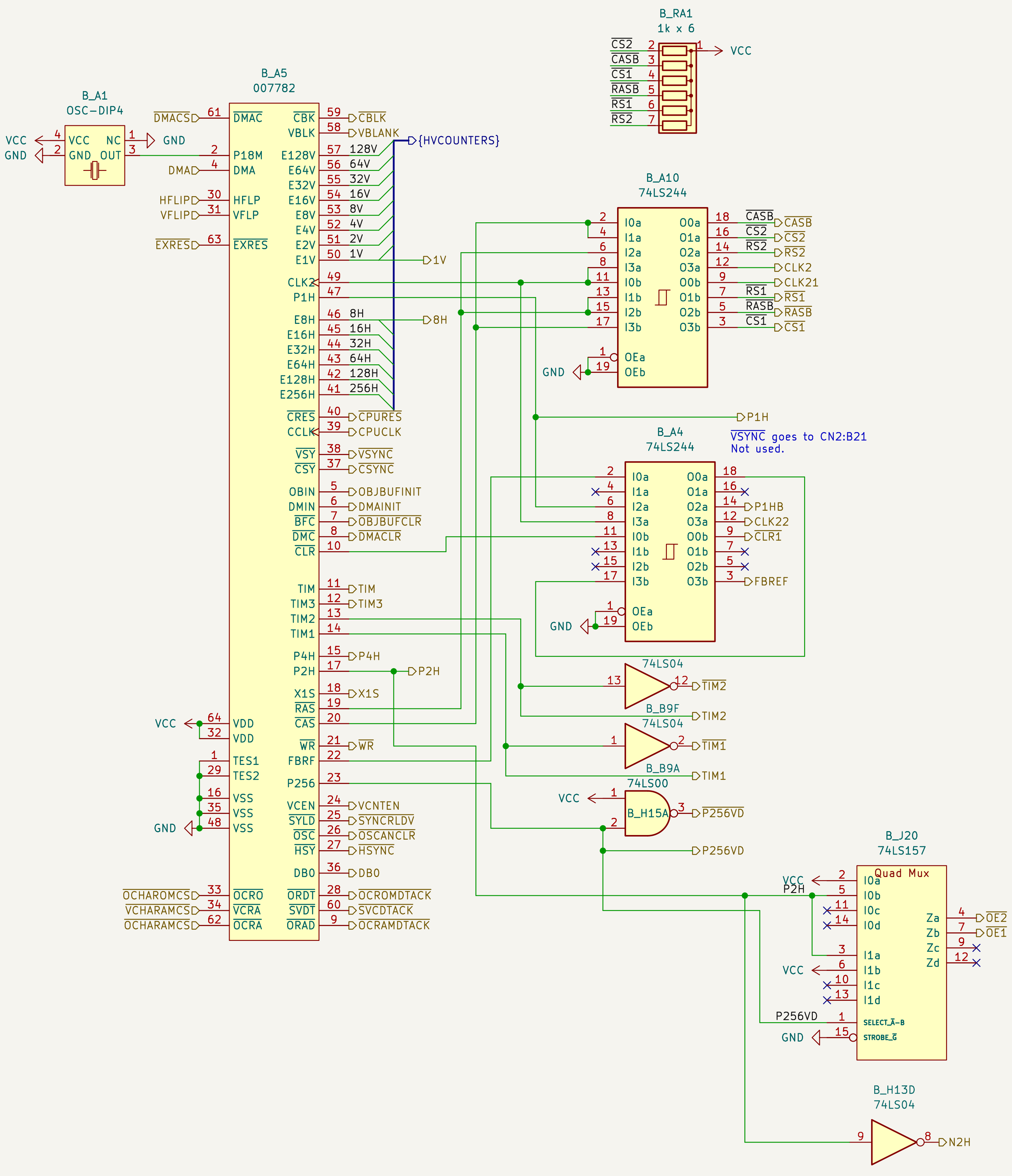
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

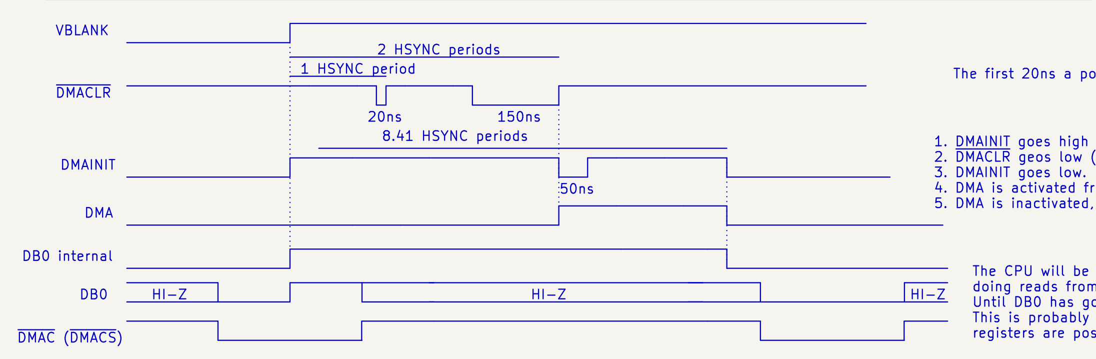
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16 - Rev B

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 4/20



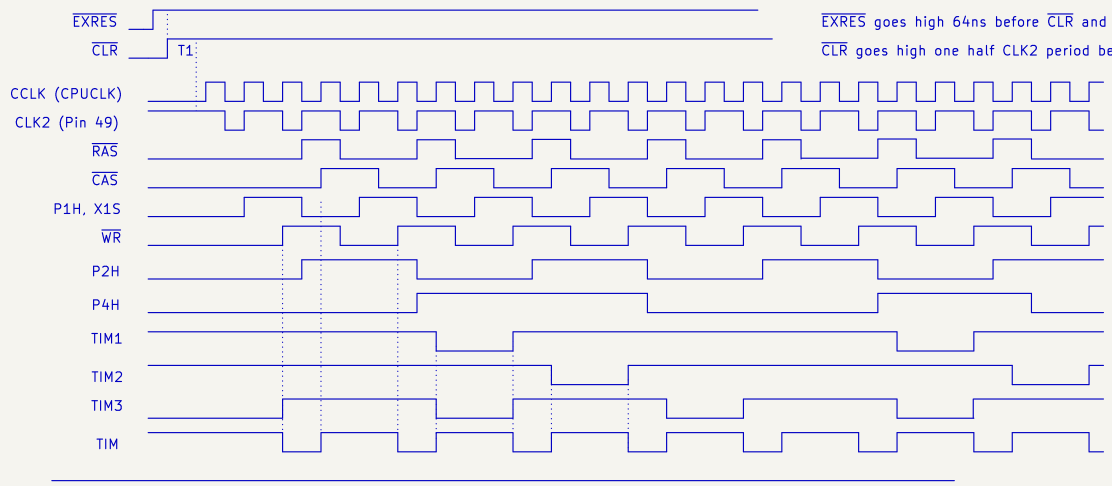
DMA Synchronization



The first 20ns a possible glitch?

1. DMAINIT goes high after VBLANK goes active
2. DMACLR geos low (the second time) and high after 150ns.
3. DMAINIT goes low.
4. DMA is activated from the 00783(Sprite engine).
5. DMA is inactivated, DMAINIT, DBO internal goes low.

The CPU will be activating regularly DMACS,
doing reads from the system registers periodically.
Until DBO has gone from low to high and low again.
This is probably to see if writes to the sprite controller
registers are possible.



EXRES goes high 64ns before CLR and stays high.

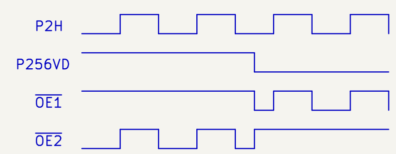
CLR goes high one half CLK2 period before the first falling edge of CLK2.

CCLK = 18.432MHz / 2 = 9.216MHz
CLK2 = 18.432MHz / 3 = 6.144MHz,
T = 1/f = 162.76ns, T = 2 x T1
f = CLK2 / 2 = 3.072MHz, T2 = 3 x T1

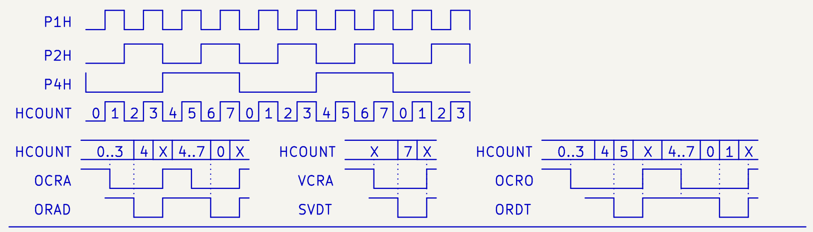
f = CLK2 / 2 = 3.072MHz
f = CLK2 / 2 = 3.072MHz
f = CLK2 / 4 = 1.536MHz
f = CLK2 / 8 = 768kHz
f = CLK2 / 8 = 768kHz
f = CLK2 / 4 = 768kHz
f = CLK2 / 4 = 1.536MHz
f = CLK2 / 2 = 3.072MHz

Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active



Chip select signals and Data Acknowledge signals



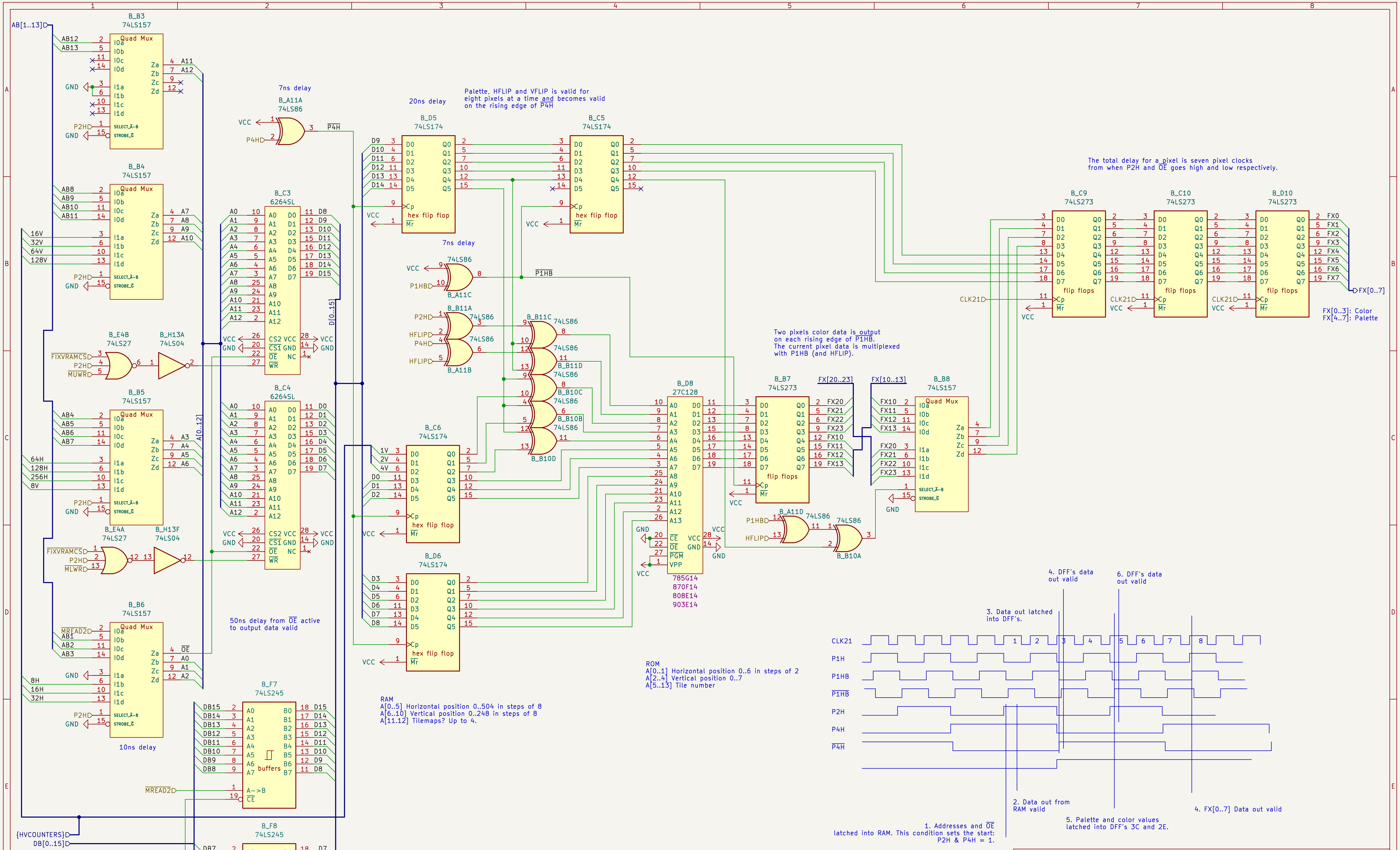
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16 — Rev B

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 5/20



Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

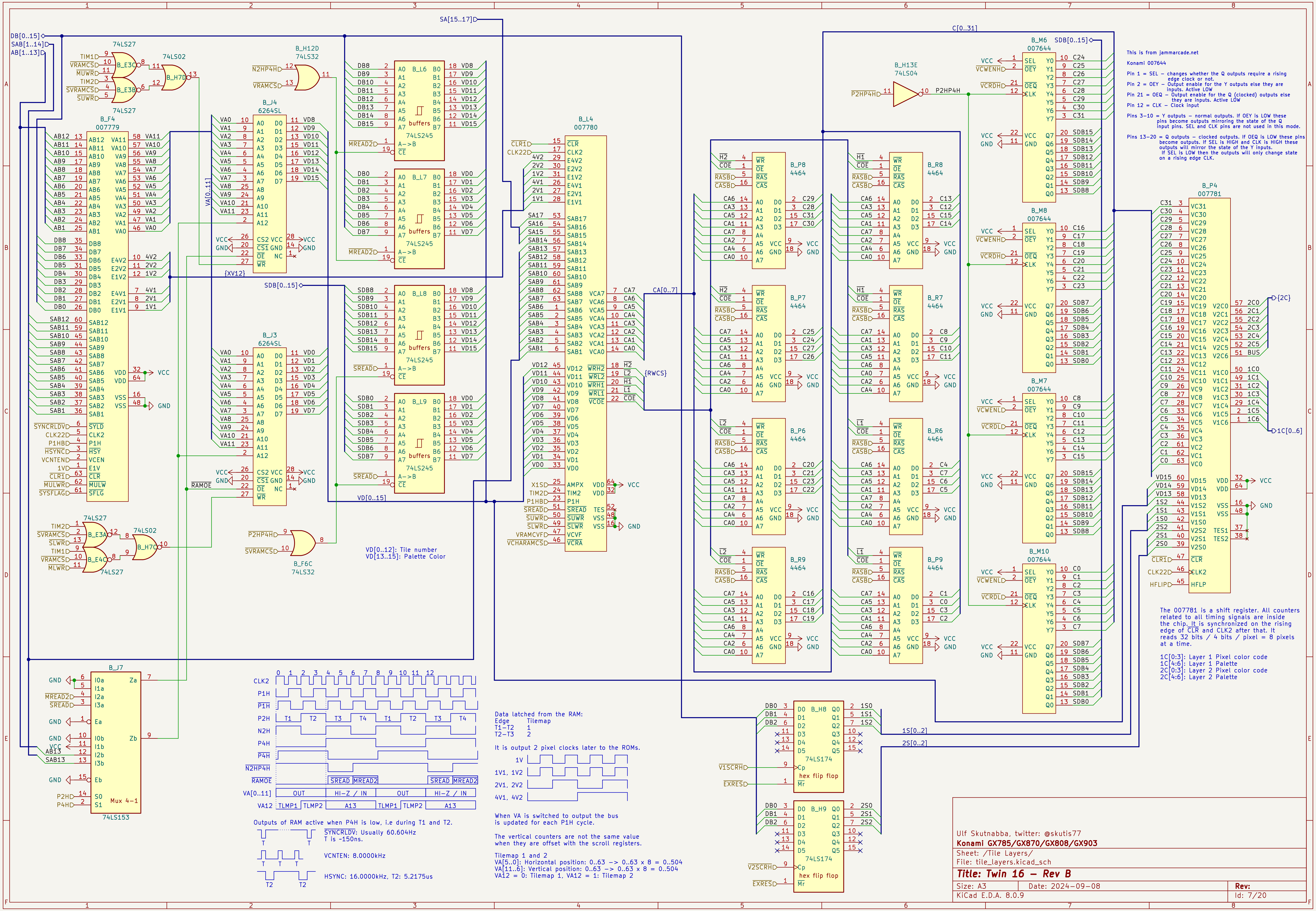
Sheet: /Fixed Layer/

File: fixed_layer.kicad_sch

Title: Twin 16 — Rev B

Size: A3 | Date: 2024-09-08 | Rev:

KiCad E.D.A. 8.0.9 | Id: 6/20



Ulf Skutnabba, twitter: @skutis77

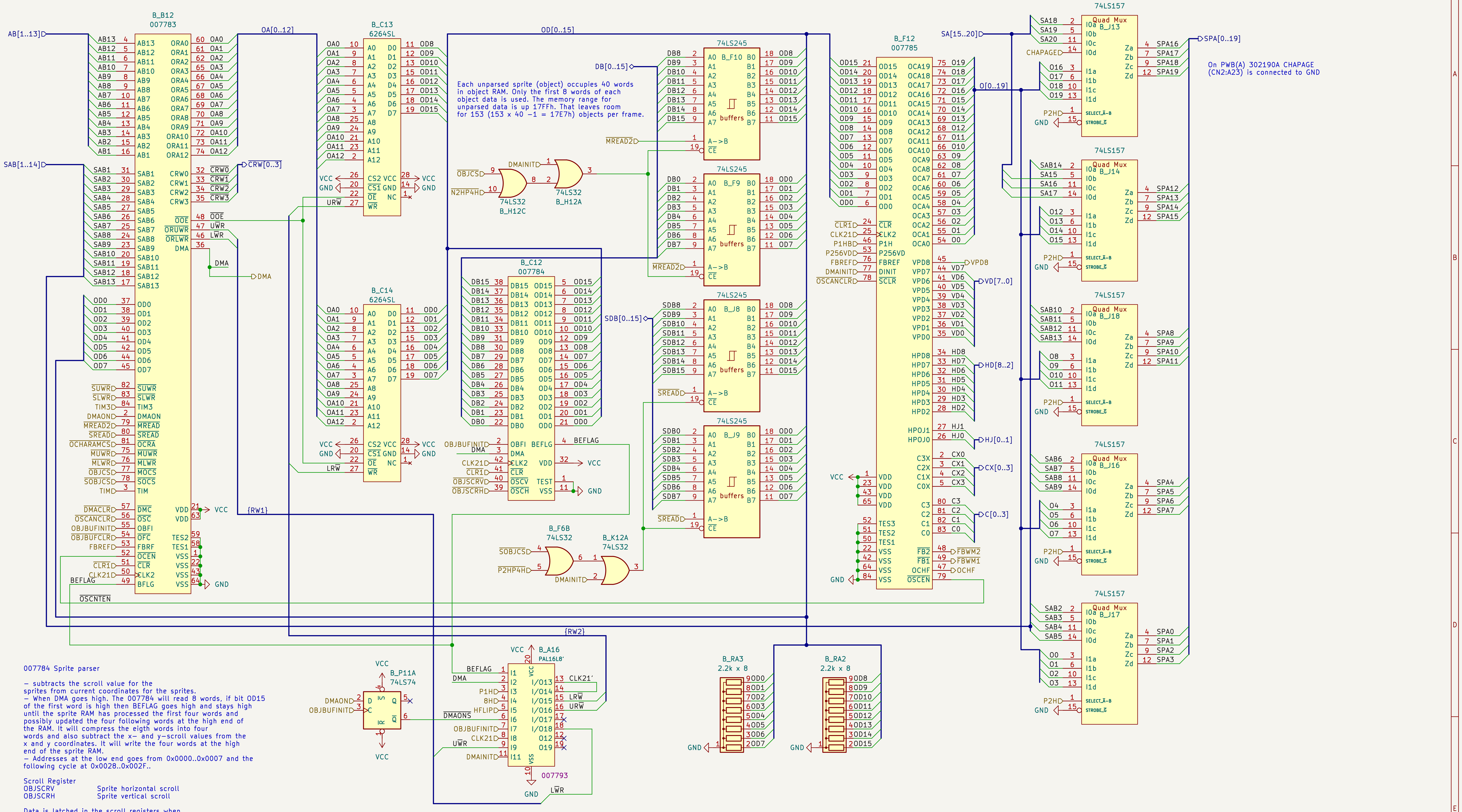
Konami GX785/GX870/GX808/GX903

Sheet: /Tile Layers/

File: tile_layers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 7/20



007784 Sprite parser

- subtracts the scroll value for the sprites from current coordinates for the sprites.
- When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x- and y-scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
- Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register

OBJSCRV	Sprite horizontal scroll
OBJSCRH	Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

```
CLK21' = CLK21

LRW = ~(DMAONS & OBJBUFINIT & CLK21' |
        BEFLAG & DMA & P1H & DMAONS & CLK21' & (~8H & HFLIP | 8H & ~HFLIP)
       ~LWR & ~DMAINIT)

URW = ~(DMAONS & OBJBUFINIT & CLK21' |
        BEFLAG & DMA & P1H & DMAONS & CLK21' & (8H & ~HFLIP | ~8H & HFLIP) |
       ~UWR & ~DMAINIT)
```

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

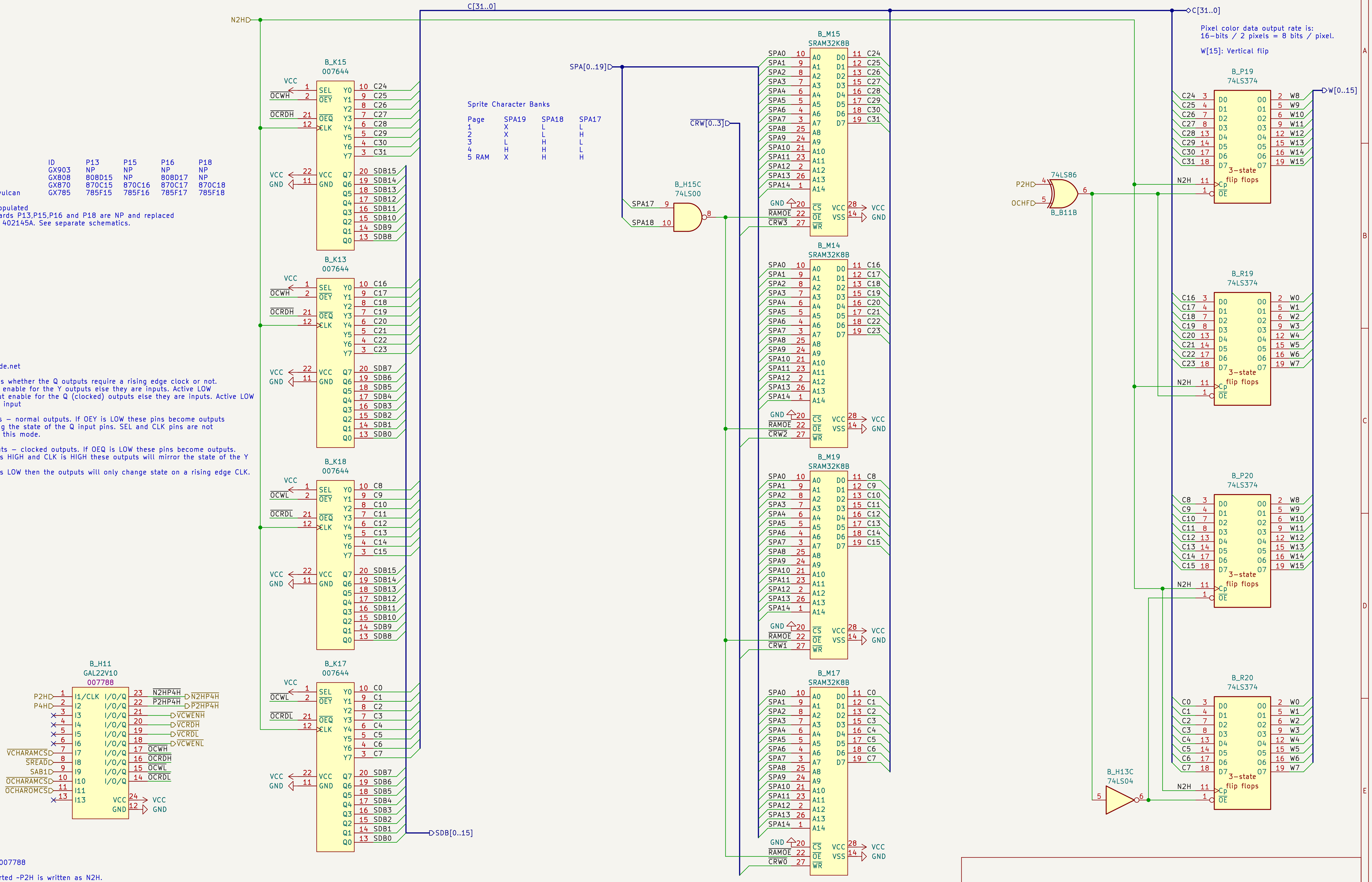
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903
Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16 - Rev B

Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 8/20

Graphics ROM and RAM



Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	H
3	L	H	L
4	H	H	L
5 RAM	X	H	H

ROM naming

	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

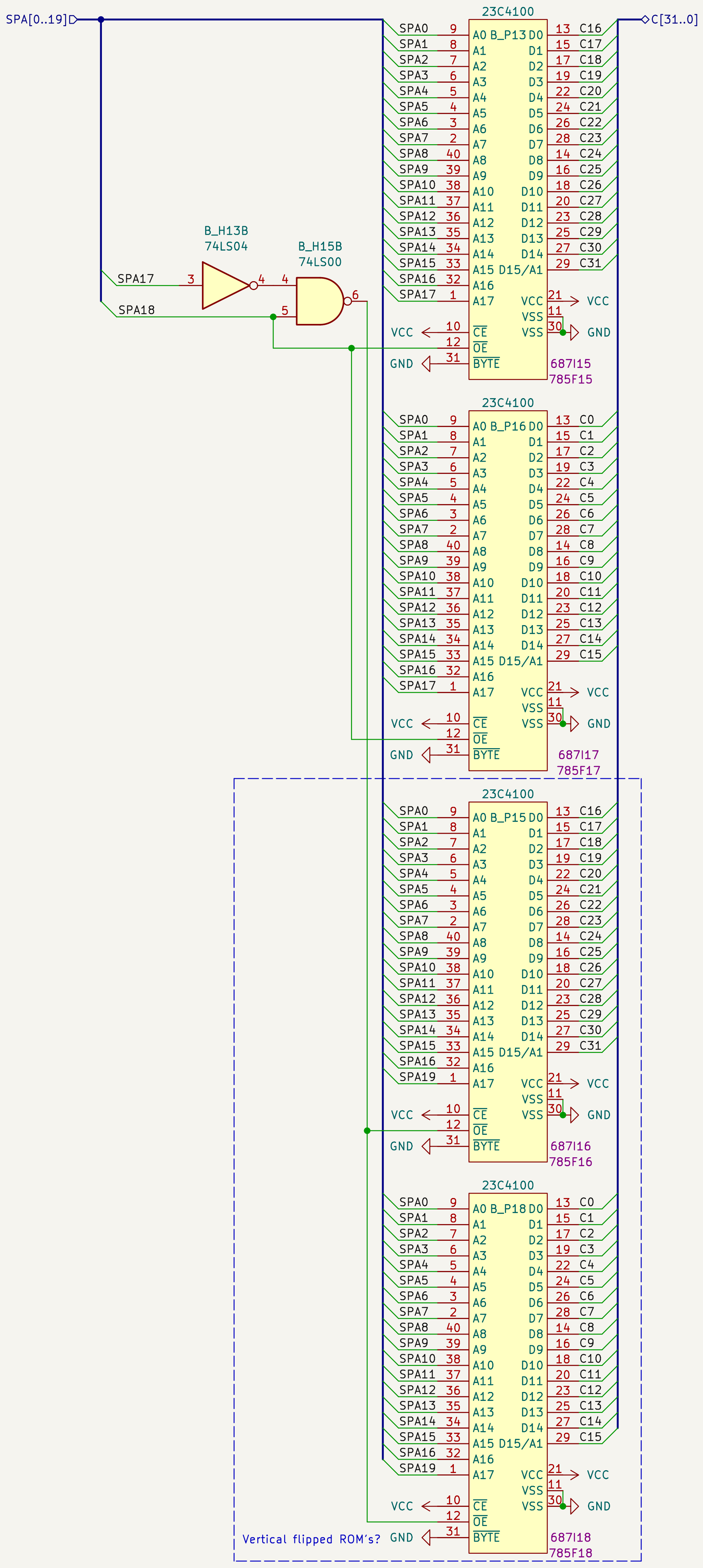
The inverted ~P2H is written as N2H.

OCRDL = ~(~SREAD & SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & SAB1 & ~OCHAROMCS)
OCWL = ~(~P2H & SREAD & SAB1 & ~OCHARAMCS)
OCRDH = ~(~SREAD & ~SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & ~SAB1 & ~OCHAROMCS)
OCWH = ~(~P2H & SREAD & ~SAB1 & ~OCHARAMCS)
VCWENL = ~(P2H & P4H & ~VCHARAMCS & SREAD & SAB1)
VCRDL = ~(~VCHARAMCS & ~SREAD & SAB1)
VCRDH = ~(~VCHARAMCS & ~SREAD & ~SAB1)
VCWENH = ~(P2H & P4H & ~VCHARAMCS & SREAD & ~SAB1)
P2HP4H = ~(P2H & P4H)
N2HP4H = ~(~P2H & P4H)

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite RAM/
File: sprite_ram.kicad_sch

Title: Twin 16 – Rev B

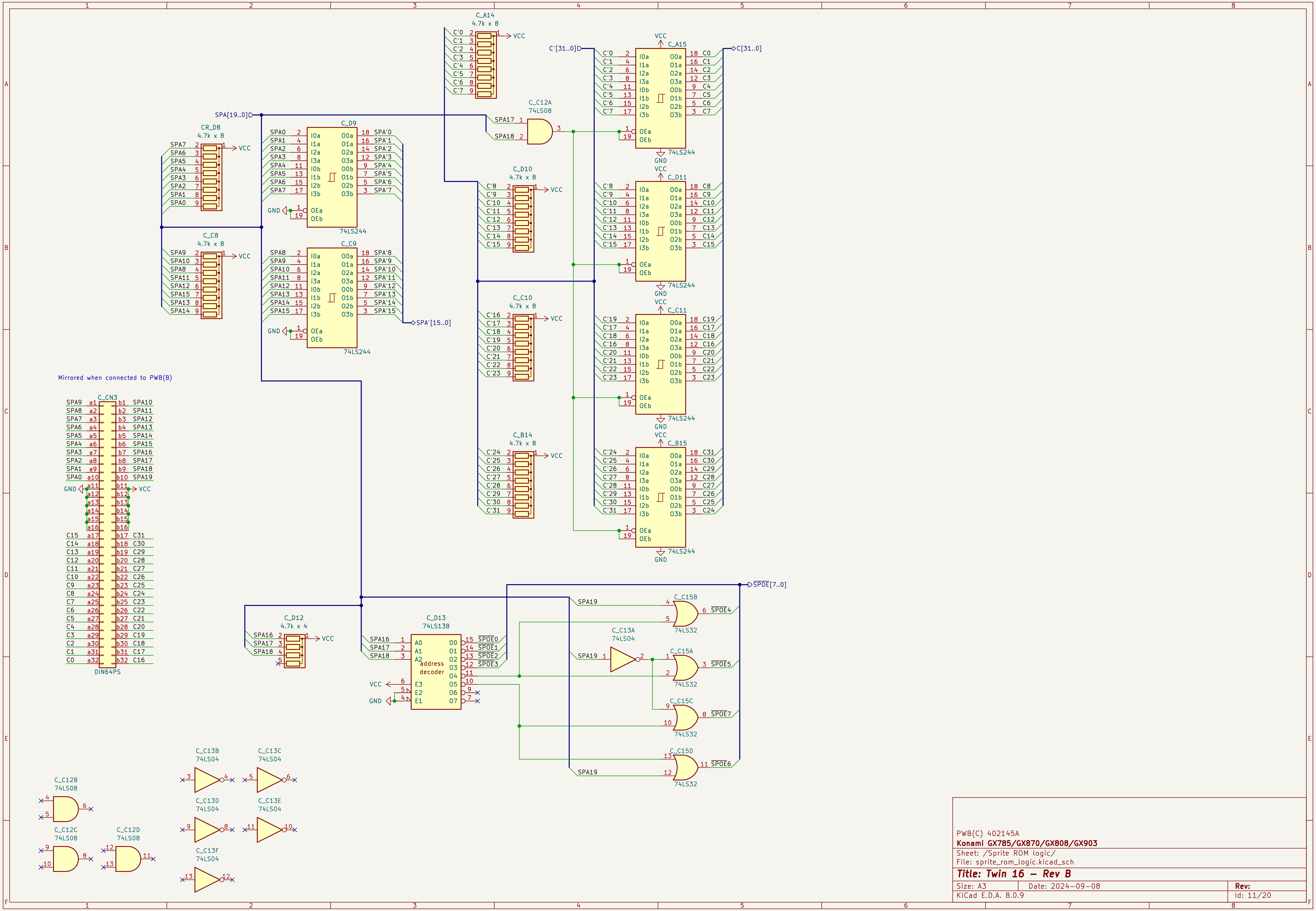
Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 9/20



Sheet: /Sprite ROM 1/
File: sprite_rom_1.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 10/20



PWB(C) 402145A

Konami GX785/GX870/GX808/GX903

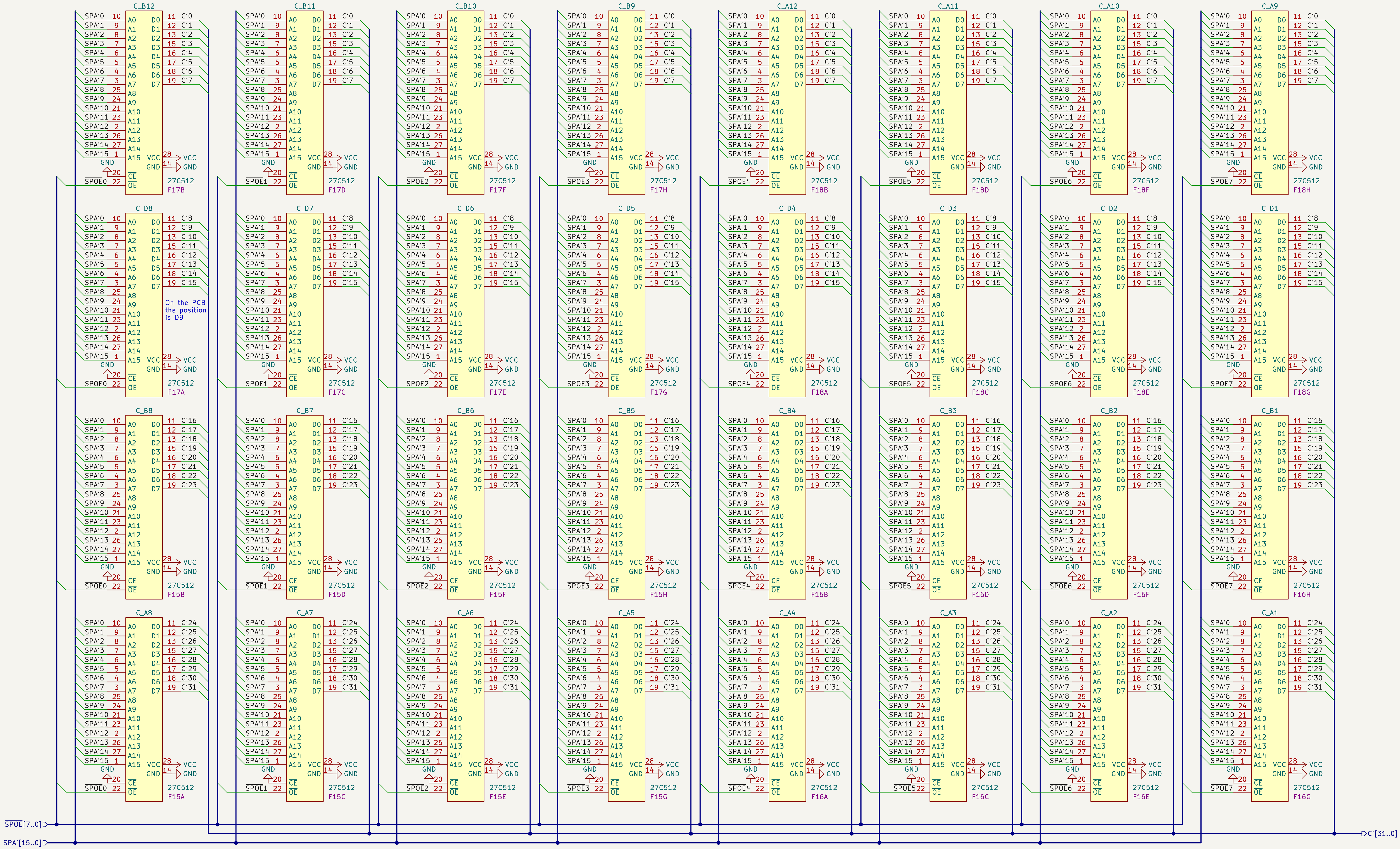
Sheet: /Sprite ROM logic/

File: sprite_rom_logic.kicad_sch

Title: Twin 16 — Rev B

Size: A3 | Date: 2024-09-08 | Rev:

KiCad E.D.A. 8.0.9 | Id: 11/20



Konami GX785/GX870/GX808/GX903

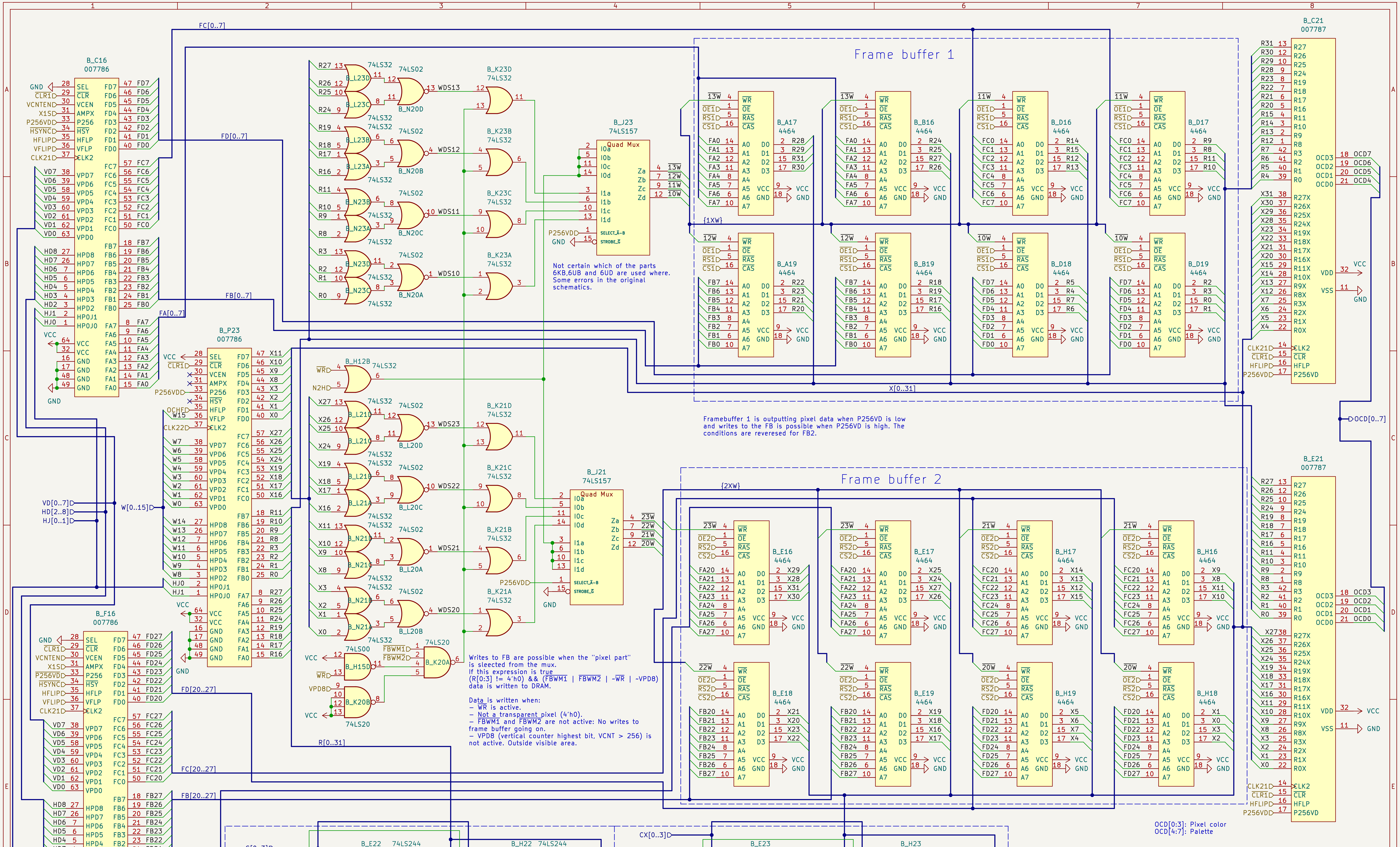
Sheet: /Sprite ROM 2/

File: sprite_rom_2.kicad_sch

Title: Twin 16 – Rev B

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 11/20

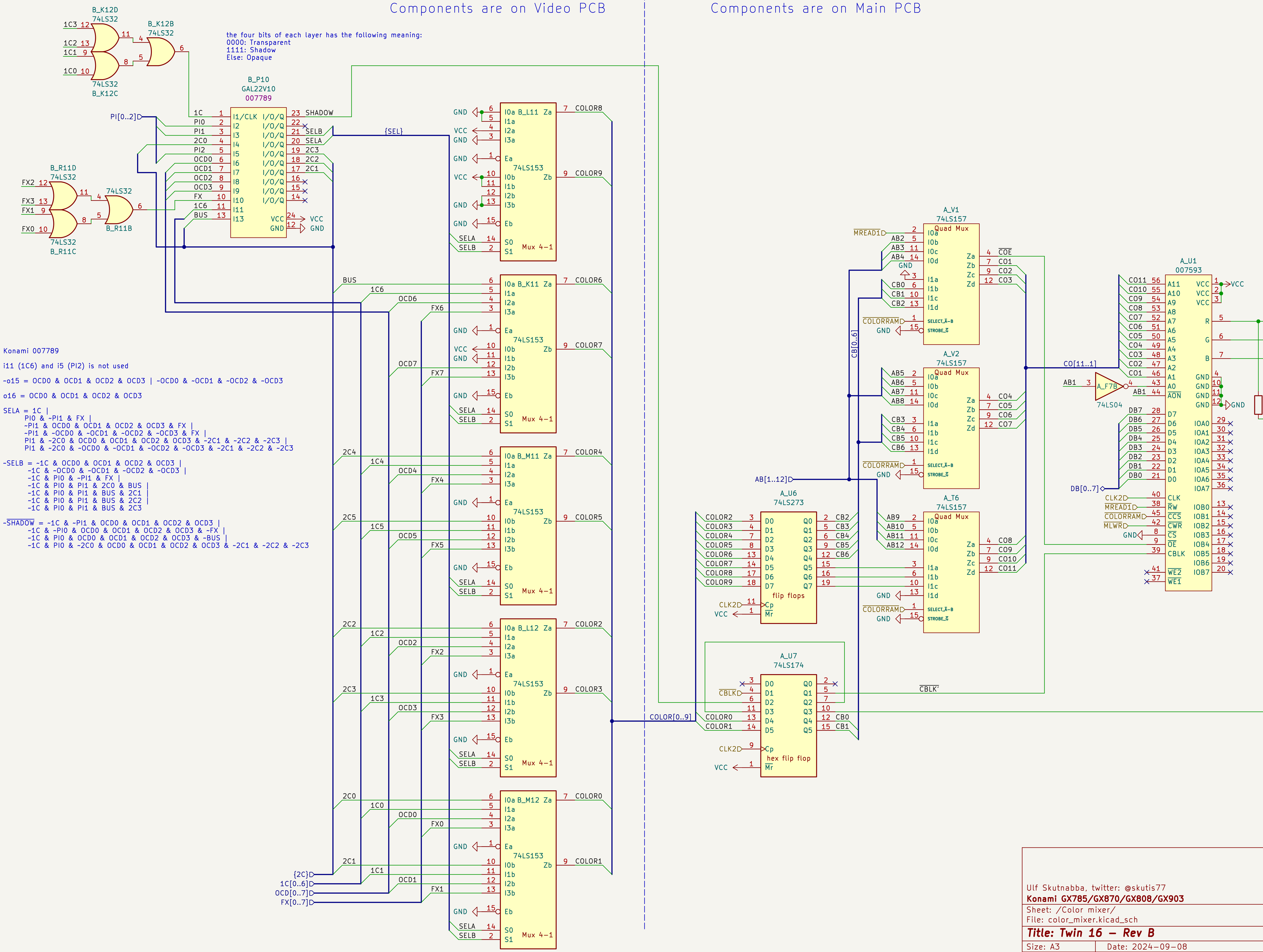


Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch
Title: Twin 16 – Rev B
Size: A3 Date: 2024-09-08 Rev:
KiCad E.D.A. 8.0.9 Id: 13/20

Components are on Video PCB

Components are on Main PCB

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque



Konami 007789

i11 (1C6) and i5 (PI2) is not used

```
~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = 1C |
      PI0 & ~PI1 & FX |
      ~PI1 & OCD0 & OCD1 & OCD2 & OCD3 & FX |
      ~PI1 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & FX |
      PI1 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3 |
      PI1 & ~2C0 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & ~2C1 & ~2C2 & ~2C3

~SELB = ~1C & OCD0 & OCD1 & OCD2 & OCD3 |
       ~1C & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 |
       ~1C & PI0 & ~PI1 & FX |
       ~1C & PI0 & PI1 & 2C0 & BUS |
       ~1C & PI0 & PI1 & BUS & 2C1 |
       ~1C & PI0 & PI1 & BUS & 2C2 |
       ~1C & PI0 & PI1 & BUS & 2C3

~SHADOW = ~1C & ~PI1 & OCD0 & OCD1 & OCD2 & OCD3 |
       ~1C & ~PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~FX |
       ~1C & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~BUS |
       ~1C & PI0 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3
```

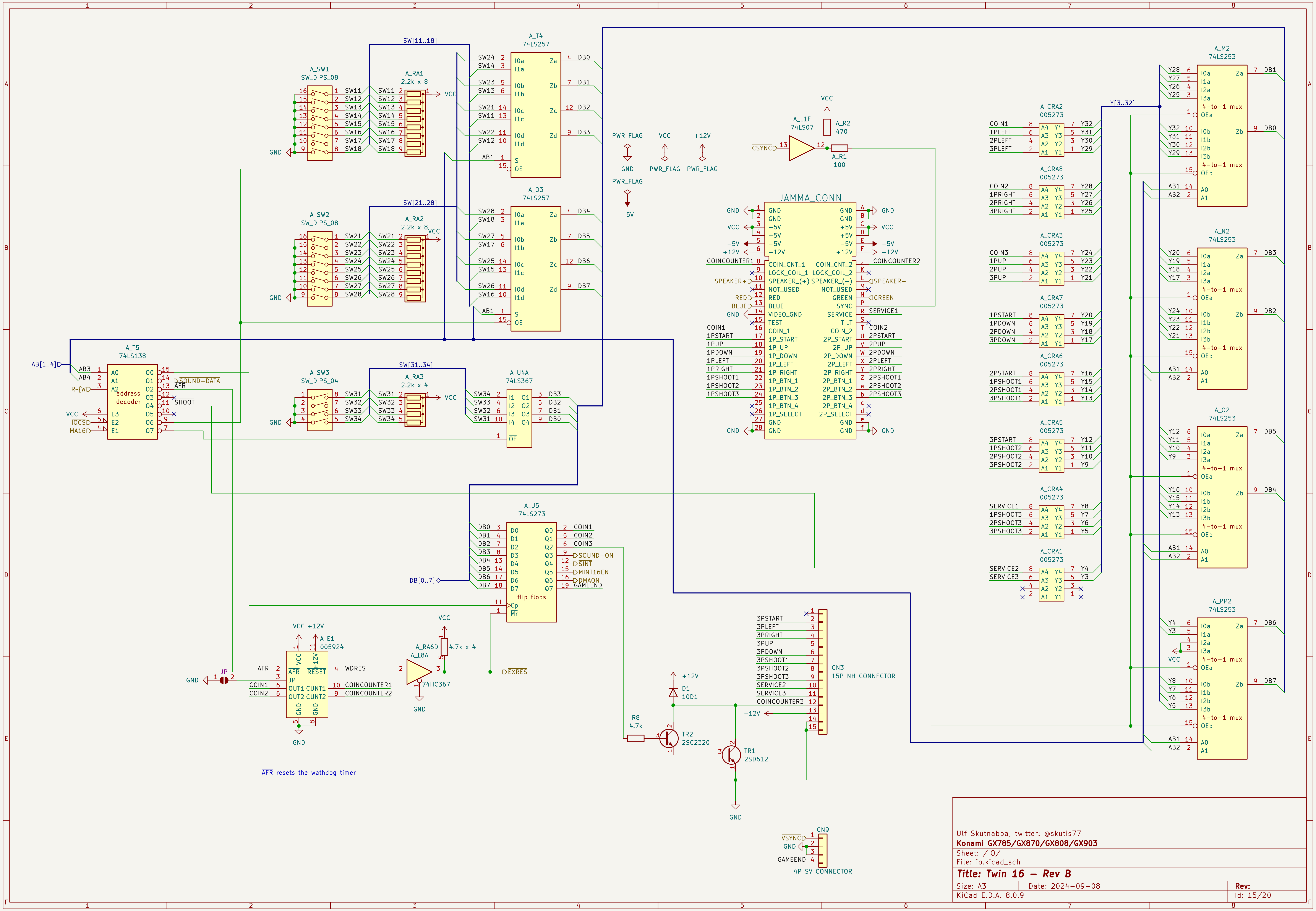

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Color mixer/
File: color_mixer.kicad_sch

Title: Twin 16 – Rev B

Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 14/20



Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

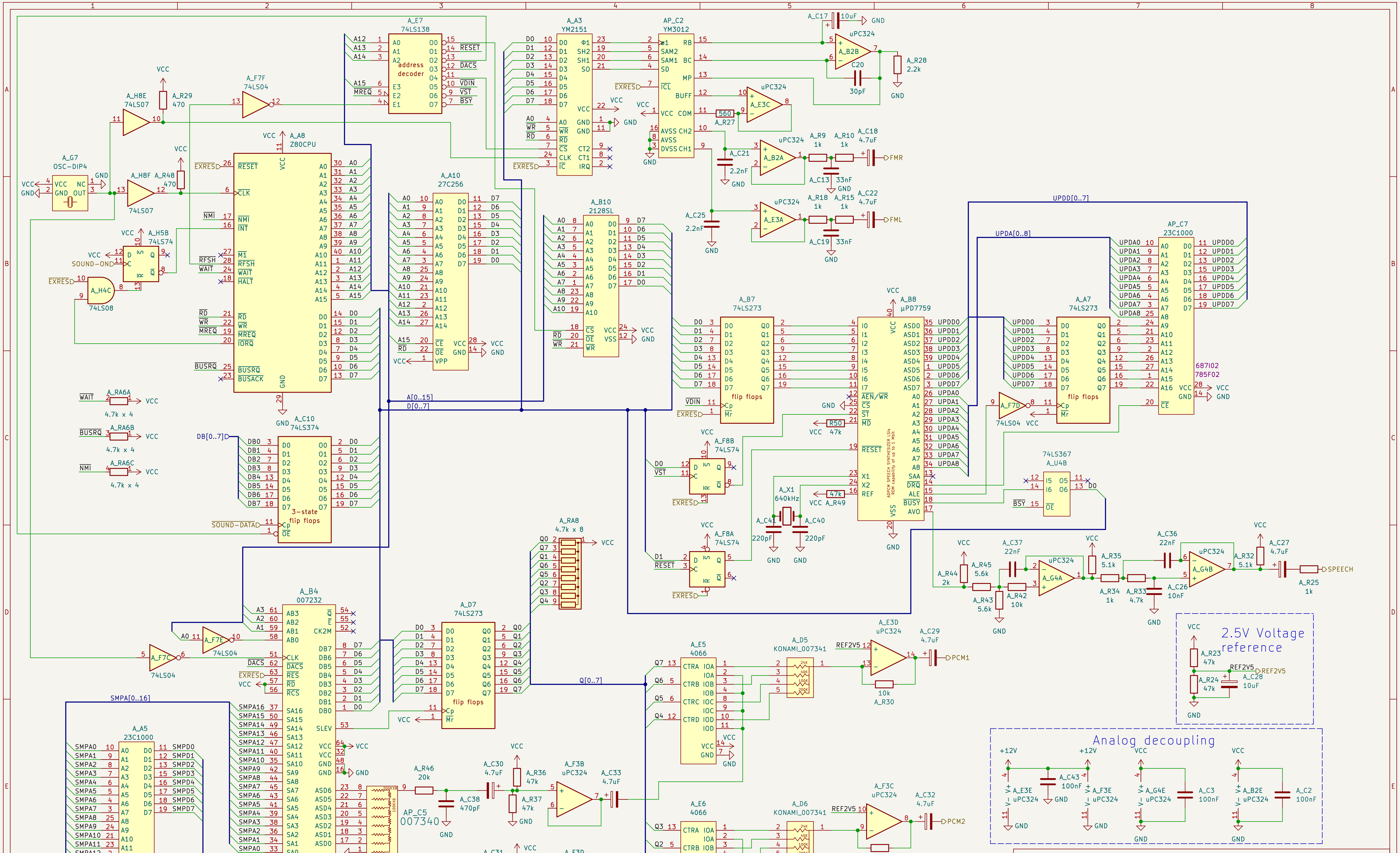
Sheet: /IO/

File: io.kicad_sch

Title: Twin 16 – Rev B

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 15/20



Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sound/

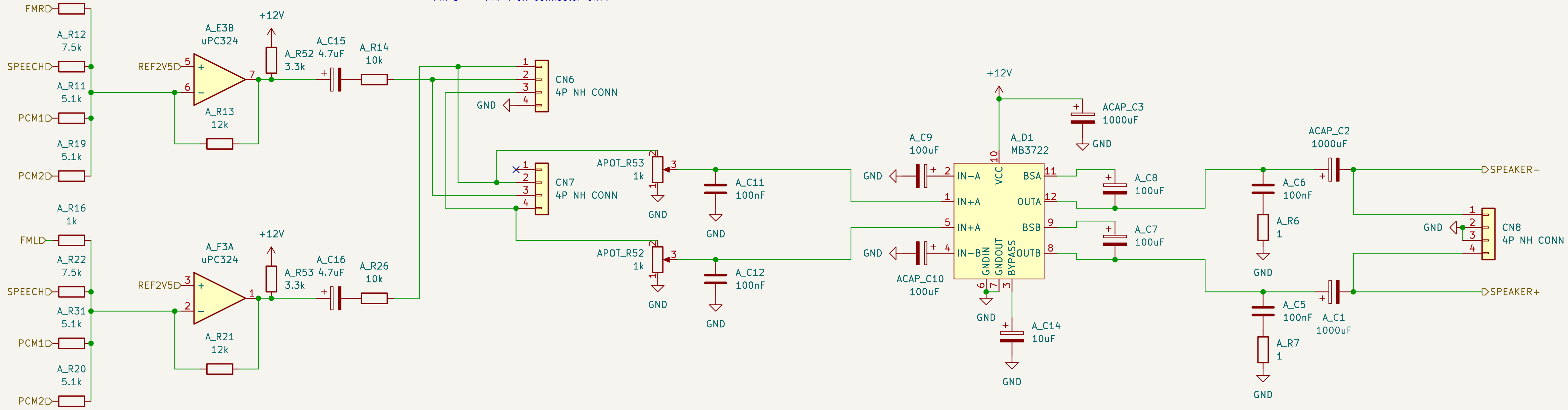
File: sound.kicad_sch

Title: Twin 16 – Rev B

Size: A3 Date: 2024-09-08 Rev:

KiCad E.D.A. 8.0.9 Id: 16/20

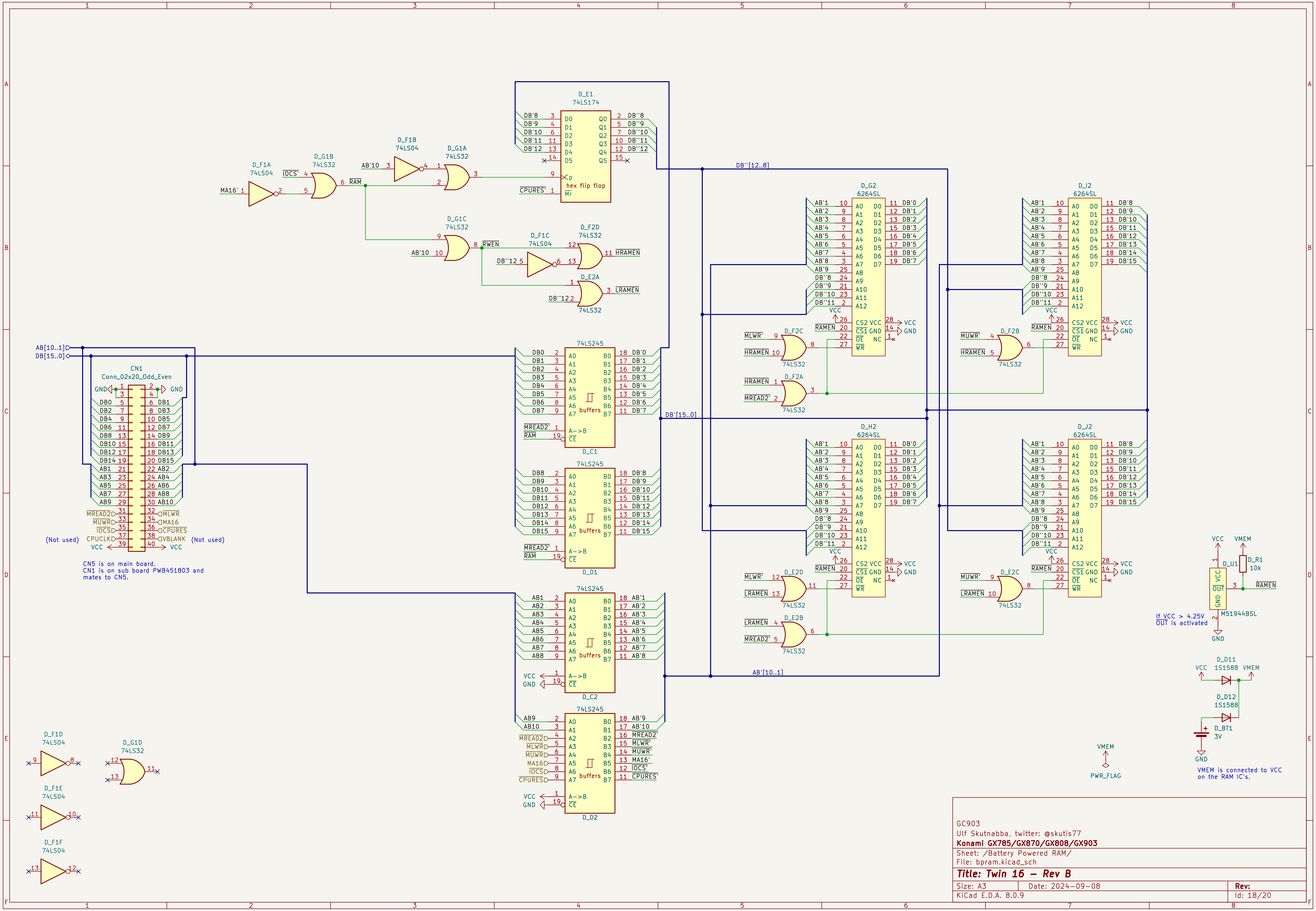
To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.



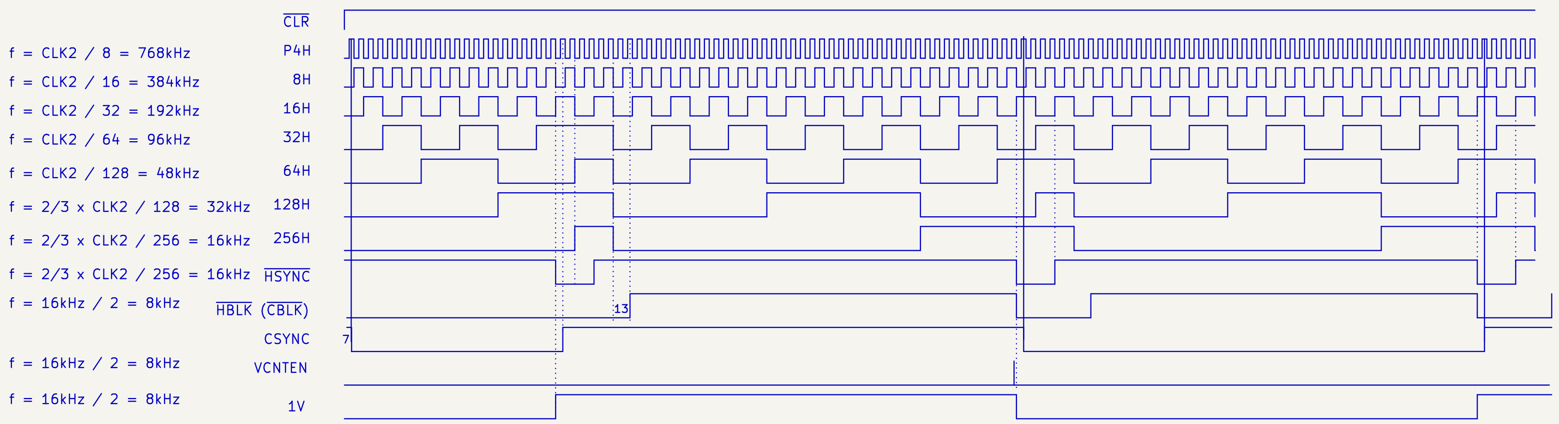
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16 - Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 17/20


D_E1
74LS174
hex flip flop
D_F1A
74LS04
D_G1B
74LS32
D_F1B
74LS04
D_G1A
74LS32
D_G1C
74LS32
D_F1C
74LS04
D_F2D
74LS32
D_E2A
74LS32
HRAMEN
LRAMEN
RWEN
RAM
DB''[12..8]
DB'[15..0]
AB'[10..1]
AB[10..1]
DB[15..0]
CN1
Conn_02x20_Odd_Even
(Not used)
(Not used)
CN5 is on main board.
CN1 is on sub board PWB451803 and
mates to CN5.
74LS245
buffers
D_C1
D_D1
D_C2
D_D2
D_G2
6264SL
D_J2
6264SL
D_H2
6264SL
D_J2
6264SL
D_F2C
74LS32
D_F2A
74LS32
D_F2B
74LS32
D_E2D
74LS32
D_E2B
74LS32
D_E2C
74LS32
D_U1
M51944BSL
D_R1
10k
If VCC > 4.25V
OUT is activated
D_D11
1S1588
D_D12
1S1588
D_BT1
3V
VMEM
PWR_FLAG
VMEM is connected to VCC
on the RAM IC's.
D_F1D
74LS04
D_G1D
74LS32
D_F1E
74LS04
D_F1F
74LS04
GC903
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Battery Powered RAM/
File: bpram.kicad_sch
Title: Twin 16 - Rev B
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 18/20

Horizontal signals

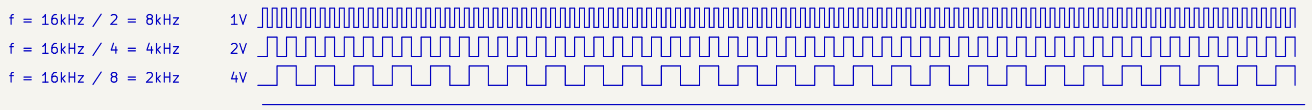


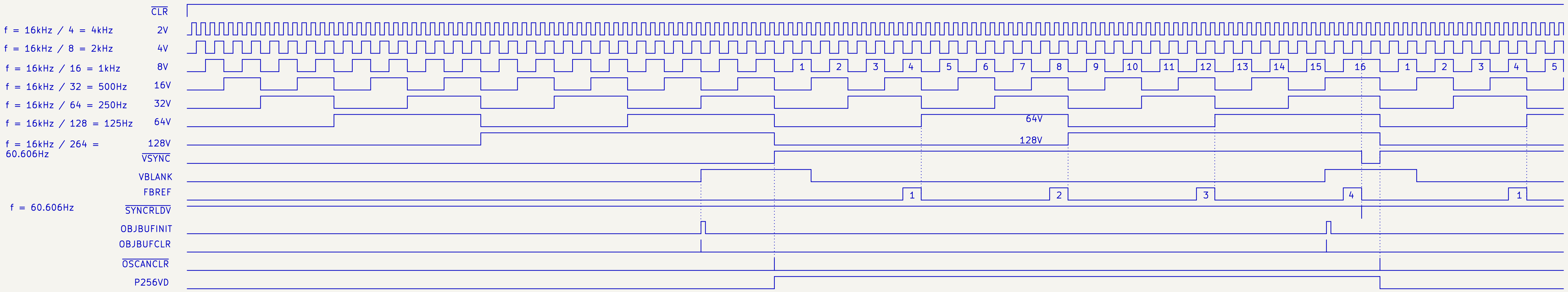
The first HSYNC is at HCOUNT 175 after reset.

– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of
HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge
of HSYNC.

Vertical signals





Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer
VBLANK High: 7 x 16 x 2 = 224, Low: 5 x 4 x 2 = 40
SYNCRLDV goes low 140ns before FBREF goes low.
SYNCRLDV goes high again when FBREF goes low.
OBJBUFCLR goes low one pixel clock before the rising edge
of VBLANK and high again when VBLANK goes high.
60.606Hz, OSCANCLR goes low for 140ns - One pixel clock cycle.
P256VD goes high/low one
P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

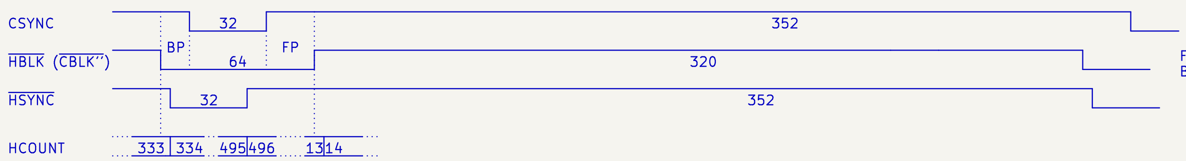
Horizontal and vertical synch timing diagrams

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

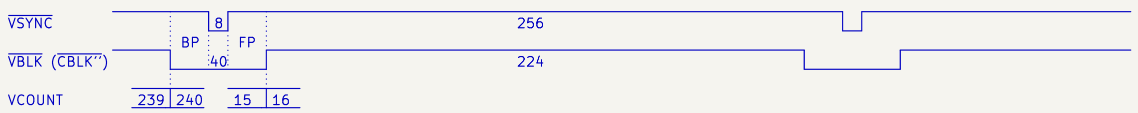
Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer 007593.
CBLK' is shifted inside the 007593 one pixel clock.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

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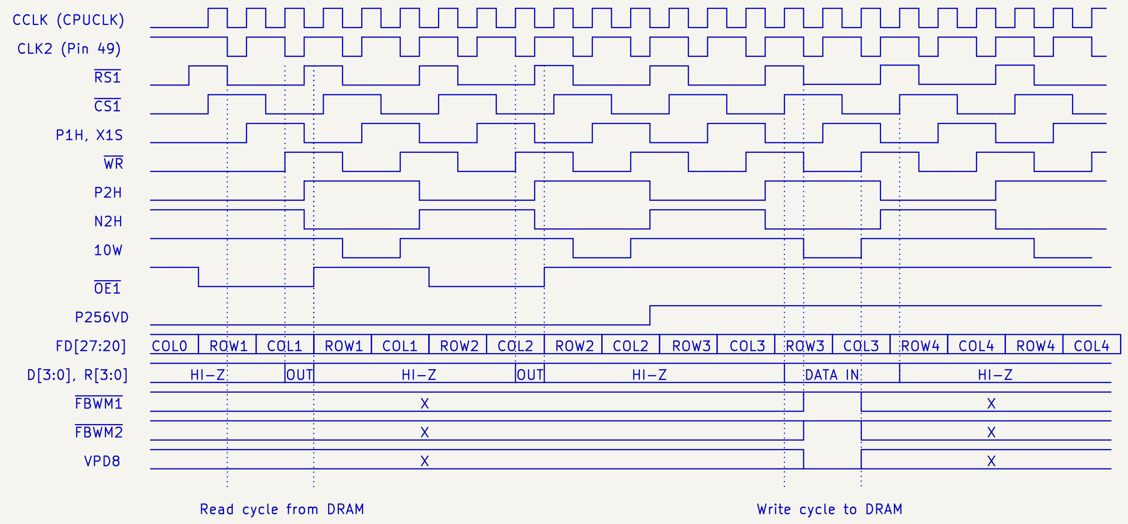

The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch
Title: Twin 16 – Rev B
Size: A3 Date: 2024-09-08 Rev:
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